TUTORIAL 13

Adjoints and Unitary Diagonalization

MOTIVATION AND GUIDING IDEA

Gram-Schmidt replaces a list of directions with an orthonormal list spanning the same space, simplifying coordinate and projection computations. Orthogonal diagonalization imposes an additional requirement: the basis vectors must also be eigenvectors. For diagonalization, therefore, begin with the eigenspaces and apply the algorithm within each one; applying it to an arbitrary basis does not generally produce eigenvectors.

NOTATION AND CONVENTIONS

- v_j is an input vector, w_j is the residual after subtracting projections, and e_i is a unit vector already obtained. Never divide by $\|w_j\|$ when $w_j = 0$.
- $\text{span}\{v_1, \ldots, v_j\}$ is the span of the input prefix; it equals the span of the output vectors obtained up to that step.
- $\mathbb{R}_n[x]$ and $\mathbb{C}_n[x]$ denote polynomials of degree at most n, including the zero polynomial. Evaluation values and integrals are computed according to the inner product specified in the question.
- $E_\lambda(T) = \ker(T - \lambda I)$ is the eigenspace belonging to λ; an eigenvector is a **nonzero** element of this space.
- T^* is the adjoint map; $A^* = \overline{A}^T$ is the conjugate transpose. The identity $[T^*]_B = [T]_B^*$ requires an orthonormal basis B.
- $T = T^*$ means self-adjointness; $TT^* = T^*T$ means normality.
- $Q^TQ = QQ^T = I$ describes a real orthogonal matrix; $U^*U = UU^* = I$ describes a complex unitary matrix. In particular, $Q^{-1} = Q^T$ and $U^{-1} = U^*$.
- In $Q^TAQ = D$ or $U^*AU = D$, column j of the change-of-basis matrix is an eigenvector belonging to d_{jj}. The column order must match the order of the eigenvalues on the diagonal.

DEFINITION 13.1

Conditions for the tools. In an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, Gram-Schmidt applies to any finite list, including a dependent list: after obtaining $e_1, \ldots, e_r$, compute

$$w_j = v_j - \sum_{i=1}^{r} \langle v_j, e_i \rangle e_i.$$

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If $w_j = 0$, skip that input vector; otherwise append $e_{r+1} = w_j/\|w_j\|$. The resulting list is an orthonormal basis of the input span. For a linearly independent input, no vector is skipped. All projection coefficients use the convention of linearity in the first argument.
If V is finite dimensional and $T : V \to V$ is linear, there is a unique linear map $T^* : V \to V$ satisfying $\langle Tv, w\rangle = \langle v, T^*w\rangle$ for all v, w. Over $\mathbb{C}$, a map is normal if and only if it has an orthonormal eigenbasis. If it is self-adjoint, all its eigenvalues are real. Over $\mathbb{R}$, every symmetric matrix admits an orthogonal diagonalization; normality alone does not suffice for a real orthogonal diagonalization. For example,

$$R = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}, \qquad RR^T = R^TR = 4I,$$

but $R^2 = -4I$, so R has no real eigenvector. When constructing an orthonormal eigenbasis, apply Gram-Schmidt **within each eigenspace** so that the linear combinations remain in that eigenspace.

13.1 Adjoints, Normality, and Diagonalization in an Orthonormal Basis

DEFINITION 13.2

Let V be a finite-dimensional inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $T : V \to V$ be linear. The **adjoint** T^* is the unique linear map satisfying

$$\langle Tv, w\rangle = \langle v, T^*w\rangle \qquad \text{for all } v, w \in V.$$

In an orthonormal basis, if A represents T, then $A^* = \overline{A}^T$ represents T^*. We call T **self-adjoint** if $T^* = T$, and **normal** if $TT^* = T^*T$. A matrix $U \in M_n(\mathbb{C})$ is **unitary** if $U^*U = UU^* = I$; its columns form an orthonormal basis of $\mathbb{C}^n$. Over the reals, a square matrix Q is **orthogonal** if $Q^TQ = QQ^T = I$.

PROPOSITION 13.1

On a finite-dimensional complex inner product space, every self-adjoint map has an orthonormal eigenbasis, and all its eigenvalues are real. This is the spectral theorem for self-adjoint maps. Thus for every Hermitian matrix $A = A^*$, there are a unitary matrix U and a real diagonal matrix D such that

$$U^*AU = D, \qquad A = UDU^*.$$

Column j of U is an eigenvector of A belonging to d_{jj}. Over $\mathbb{R}$, every real symmetric matrix $A = A^T$ has a real orthonormal eigenbasis; hence there is an orthogonal matrix Q such that

$$Q^TAQ = D, \qquad A = QDQ^T.$$

Diagonalization by an orthogonal matrix is called **orthogonal diagonalization**; over the complex numbers, we use the term **unitary diagonalization**.

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RECALL AND TOOLS

To find an orthogonal diagonalization of a real symmetric matrix, find a basis of each eigenspace, apply Gram-Schmidt within each eigenspace as needed, and normalize. Eigenvectors of a symmetric matrix belonging to distinct eigenvalues are already orthogonal. Place the unit vectors in the columns of Q, and place the corresponding eigenvalues in the same order on the diagonal of D. Over the complex numbers, the same construction applies to a Hermitian matrix, using U^* in place of Q^T.

EXERCISE 13.1

Let V be a finite-dimensional complex inner product space, and let $T : V \to V$ be linear. Suppose every eigenvector of $T + T^*$ is also an eigenvector of $T - T^*$. Please prove that T is normal.

WORKED SOLUTION

Write

$$H := T + T^*, \qquad K := T - T^*.$$

We aim to find an orthonormal basis in which both maps act by scalar multiplication on every basis vector. First compute the adjoints:

$$H^* = (T + T^*)^* = T^* + (T^*)^* = T^* + T = H,$$

and

$$K^* = (T - T^*)^* = T^* - (T^*)^* = T^* - T = -K.$$

Thus H is self-adjoint. If $V = \{0\}$, the result is immediate. Otherwise, the spectral theorem provides an orthonormal basis $B = (e_1, \ldots, e_n)$ and real numbers $h_1, \ldots, h_n$ such that

$$He_j = h_j e_j \qquad (1 \le j \le n).$$

Every e_j is an eigenvector of H, so the assumption provides $k_j \in \mathbb{C}$ such that

$$Ke_j = k_j e_j.$$

The number k_j may depend on j; we need not assume that the eigenvalues of H are distinct. In the basis B, both matrices are diagonal:

$$[H]_B = D_H := \operatorname{diag}(h_1, \ldots, h_n), \qquad [K]_B = D_K := \operatorname{diag}(k_1, \ldots, k_n).$$

For every j,

$$HKe_j = H(k_j e_j) = k_j h_j e_j, \qquad KHe_j = K(h_j e_j) = h_j k_j e_j.$$

Scalars commute, so $HKe_j = KHe_j$ for every basis vector. Hence $HK = KH$ on all of V. Now recover T and T^*:

$$H + K = 2T, \qquad H - K = 2T^*, \qquad T = \frac{H + K}{2}, \quad T^* = \frac{H - K}{2}.$$

Compute the two products separately, without assuming in advance that the factors com-

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EXERCISE 13.1 CONTINUED

mute:

$$\begin{aligned} TT^* &= \frac{1}{4}(H+K)(H-K) \\ &= \frac{1}{4}\big(HH - HK + KH - KK\big) \\ &= \frac{1}{4}\big(H^2 - HK + KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2), \end{aligned}$$

where the last line uses $HK = KH$. Similarly,

$$\begin{aligned} T^*T &= \frac{1}{4}(H-K)(H+K) \\ &= \frac{1}{4}\big(HH + HK - KH - KK\big) \\ &= \frac{1}{4}\big(H^2 + HK - KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2). \end{aligned}$$

Thus $TT^* = T^*T$, so T is normal.

We also make explicit the connection with unitary diagonalization and the order of the matrices. First choose a fixed orthonormal basis of V, write $A = [T]$ in that basis, and form U from the coordinate columns of $e_1, \dots, e_n$ in the same basis. Then U is unitary, and

$$U^*(A + A^*)U = D_H, \qquad U^*(A - A^*)U = D_K.$$

Adding the identities and dividing by 2 gives

$$U^*AU = D := \frac{1}{2}(D_H + D_K), \qquad A = UDU^*.$$

Thus $[T]_B = D$ and $[T^*]_B = D^*$. For diagonal matrices,

$$(DD^*)_{ij} = \sum_{\ell=1}^{n} d_{i\ell}\overline{d_{j\ell}} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j, \end{cases}$$

and

$$(D^*D)_{ij} = \sum_{\ell=1}^{n} \overline{d_{\ell i}}\, d_{\ell j} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j. \end{cases}$$

We can therefore check the equality again in matrix form:

$$\begin{aligned} AA^* &= (UDU^*)(UD^*U^*) = UD(U^*U)D^*U^* = U(DD^*)U^*, \\ A^*A &= (UD^*U^*)(UDU^*) = UD^*(U^*U)DU^* = U(D^*D)U^*. \end{aligned}$$

The two results agree. The basis that diagonalizes the self-adjoint part also diagonalizes the other part by the hypothesis, and consequently diagonalizes T itself.

All wrapped up. Victory!

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EXERCISE 13.2

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $A \in M_n(\mathbb{F})$. For any matrix C, please write $C^* := \overline{C}^T$; over $\mathbb{R}$, $C^* = C^T$. Following the convention that the inner product is linear in its first argument, define

$$\langle x, y \rangle := y^* x = x^T \overline{y}.$$

1. Please prove that $\langle Av, w \rangle = \langle v, A^* w \rangle$ for all $v, w \in \mathbb{F}^n$.
2. Let $G \in M_n(\mathbb{F})$ be Hermitian and positive definite. Define

   $$\langle x, y \rangle_G := y^* G x.$$

   Please prove that this is an inner product, and please show that there is a unique matrix A^{*_G} satisfying

   $$\langle Av, w \rangle_G = \langle v, A^{*_G} w \rangle_G \quad \text{for all } v, w \in \mathbb{F}^n,$$

   and that $A^{*_G} = G^{-1} A^* G$.

WORKED SOLUTION

1. The identity $(A^* w)^* = w^* A$ gives

   $$\langle Av, w \rangle = w^* A v = (A^* w)^* v = \langle v, A^* w \rangle.$$

2. Linearity in the first argument follows from the linearity of matrix multiplication. Since $G^* = G$,

   $$\overline{\langle y, x \rangle_G} = \overline{x^* G y} = (x^* G y)^* = y^* G^* x = y^* G x = \langle x, y \rangle_G.$$

   Furthermore, $\langle x, x \rangle_G = x^* G x > 0$ for every $x \neq 0$, and $\langle 0, 0 \rangle_G = 0$. The matrix G is invertible: otherwise $Gx = 0$ for some $x \neq 0$ would imply $x^* G x = 0$. Set $B := G^{-1} A^* G$. Then

   $$B^* = (G^{-1} A^* G)^* = G A G^{-1},$$

   and hence

   $$\langle v, Bw \rangle_G = (Bw)^* G v = w^* B^* G v = w^* G A v = \langle Av, w \rangle_G.$$

   For uniqueness, if C also satisfies the identity, then $w^*(C^* G - G A)v = 0$ for all v, w. Choosing $v = e_j$ and $w = e_i$ shows that every entry of $C^* G - G A$ is zero. Thus

   $$C^* G = G A \implies C^* = G A G^{-1} \implies C = G^{-1} A^* G = B.$$

We've reached the finish. Victory!

13.2 Linear Maps on Inner Product Spaces

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EXERCISE 13.3

Let $(V, \langle \cdot, \cdot \rangle)$ be a real inner product space of dimension $n \geq 3$. Let $v, w \in V$ be orthonormal: that is, $\langle v, v \rangle = \langle w, w \rangle = 1$ and $\langle v, w \rangle = 0$. Define the linear map $T : V \to V$ by

$$T(u) = \langle u, v \rangle \, w + \langle u, w \rangle \, v.$$

1. Please find all eigenvalues of T, all eigenvectors, and the characteristic polynomial of T.
2. Please prove that $\|T(u)\| \leq \|u\|$ for every $u \in V$, and please find all u for which equality holds.

WORKED SOLUTION

1. Start with the eigenvalue equation and compare components using an orthogonal decomposition:

   $$T(u) = \lambda u.$$

   Decompose each $u \in V$ using the orthogonal direct sum

   $$V = \text{span}\{v, w\} \oplus (\text{span}\{v, w\})^{\perp}.$$

   Here $\{v, w\}^{\perp}$ denotes the orthogonal complement of $\text{span}\{v, w\}$. Thus there are unique $\alpha, \beta \in \mathbb{R}$ and $y \in \{v, w\}^{\perp}$ such that

   $$u = \alpha v + \beta w + y.$$

   Now compute $T(u)$: since $y \perp v, w$, we have $\langle y, v \rangle = \langle y, w \rangle = 0$, so

   $$\langle u, v \rangle = \langle \alpha v + \beta w + y, \ v \rangle = \alpha, \qquad \langle u, w \rangle = \langle \alpha v + \beta w + y, \ w \rangle = \beta.$$

   Hence

   $$T(u) = \alpha w + \beta v.$$

   Substitute into the eigenvalue equation $T(u) = \lambda u$ to obtain

   $$\alpha w + \beta v = \lambda(\alpha v + \beta w + y).$$

   Compare components in the direct sum $\text{span}\{v, w\} \oplus \{v, w\}^{\perp}$. The component in $\{v, w\}^{\perp}$ on the left is zero, so necessarily

   $$\lambda y = 0.$$

   Consequently:

   - If $\lambda \neq 0$, then $y = 0$, so $u \in \text{span}\{v, w\}$.
   - If $\lambda = 0$, then $T(u) = 0$, and $\alpha w + \beta v = 0$ gives $\alpha = \beta = 0$, so $u = y \in \{v, w\}^{\perp}$.

   Now consider $\lambda \neq 0$. Then $u = \alpha v + \beta w$ and $y = 0$. The eigenvalue equation becomes

   $$\alpha w + \beta v = \lambda(\alpha v + \beta w).$$

   Comparing coefficients of v and w gives the system

   $$\beta = \lambda \alpha, \qquad \alpha = \lambda \beta.$$

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EXERCISE 13.3 CONTINUED

Substitution gives $\alpha = \lambda(\lambda\alpha) = \lambda^2\alpha$, so

$$(\lambda^2 - 1)\alpha = 0.$$

If $\alpha = 0$, then $\beta = \lambda\alpha = 0$ as well, giving $u = 0$, contrary to the definition of an eigenvector. Thus $\alpha \neq 0$, and

$$\lambda^2 - 1 = 0 \quad \Longleftrightarrow \quad \lambda \in \{1, -1\}.$$

Now determine the eigenvectors:

- For $\lambda = 1$, $\beta = \alpha$, so $u = \alpha(v + w)$; hence

$$V_1 = \text{span}\{v + w\}.$$

- For $\lambda = -1$, $\beta = -\alpha$, so $u = \alpha(v - w)$; hence

$$V_{-1} = \text{span}\{v - w\}.$$

- For $\lambda = 0$, we obtained $u \in \{v, w\}^\perp$, so

$$V_0 = \{v, w\}^\perp.$$

In particular, there are no other eigenvalues, because the eigenvalue equation forced $\lambda \in \{0, 1, -1\}$.

To compute the characteristic polynomial, choose an orthonormal basis of V of the form

$$B = (v, w, e_3, \ldots, e_n),$$

where $e_3, \ldots, e_n$ is an orthonormal basis of $\{v, w\}^\perp$. Then

$$T(v) = w, \qquad T(w) = v, \qquad T(e_i) = 0 \ \ (i = 3, \ldots, n).$$

The matrix representing T relative to B is therefore

$$[T]_B = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Hence its characteristic polynomial is

$$p_T(\lambda) = \det(\lambda I - [T]_B) = \det \begin{pmatrix} \lambda & -1 & 0 & \cdots & 0 \\ -1 & \lambda & 0 & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda \end{pmatrix} = (\lambda^2 - 1)\lambda^{n-2}.$$

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EXERCISE 13.3 CONTINUED

2. Set $\alpha := \langle u, v\rangle$ and $\beta := \langle u, w\rangle$, and define

$$s := \alpha v + \beta w \in \text{span}\{v, w\}.$$

Since v, w are orthonormal,

$$\|s\|^2 = \langle s, s\rangle = \alpha^2 + \beta^2 = \langle u, v\rangle^2 + \langle u, w\rangle^2 = \|T(u)\|^2.$$

Also,

$$\langle u, s\rangle = \alpha\langle u, v\rangle + \beta\langle u, w\rangle = \alpha^2 + \beta^2 = \|s\|^2.$$

By the Cauchy-Schwarz inequality,

$$|\langle u, s\rangle| \leq \|u\|\,\|s\|.$$

Since $|\langle u, s\rangle| = \|s\|^2$, we obtain

$$\|s\|^2 \leq \|u\|\,\|s\|.$$

If $s = 0$, then $\|T(u)\| = \|s\| = 0 \leq \|u\|$. If $s \neq 0$, divide by $\|s\|$ to obtain

$$\|T(u)\| = \|s\| \leq \|u\|.$$

To check equality, write $u = s + y$ with $y \in \{v, w\}^{\perp}$. By the Pythagorean identity,

$$\|u\|^2 = \|s\|^2 + \|y\|^2 = \|T(u)\|^2 + \|y\|^2.$$

Thus $\|T(u)\| = \|u\|$ if and only if $y = 0$, or equivalently $u \in \text{span}\{v, w\}$. This argument also includes $s = 0$, in which case equality requires $u = 0$.

Yes! Victory!

EXERCISE 13.4

Let $n \geq 3$ be a natural number and let $a \in \mathbb{R}$. Define $A_n(a) \in M_n(\mathbb{R})$ by

$$(A_n(a))_{ij} = \begin{cases} 1, & i = j, \\ a, & i \neq j. \end{cases}$$

Recall the definition: a Hermitian matrix $H \in M_n(\mathbb{C})$ is positive semidefinite if $x^*Hx \geq 0$ for every $x \in \mathbb{C}^n$, and positive definite if $x^*Hx > 0$ for every $x \neq 0$.

1. For every a, please find all eigenvalues of $A_n(a)$ and their multiplicities.
2. Please determine for which values of a the matrix $A_n(a)$ is positive definite, and for which values it is positive semidefinite.
3. Please find $\text{rank}(A_n(a))$ and $\ker(A_n(a))$ in the two boundary cases where $A_n(a)$ is positive semidefinite but not positive definite.

In addition, let $B \in M_n(\mathbb{C})$ and please write $B^* = \overline{B}^T$. Define

$$H := B^*B + I.$$

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EXERCISE 13.4 CONTINUED

(4) Please prove that H is positive definite and hence invertible.

(5) Please prove that, for every $x \in \mathbb{C}^n$,

$$x^* H^{-1} x \leq \|x\|^2,$$

and also prove that

$$\det(H) \geq 1,$$

with equality if and only if $B = 0$.

WORKED SOLUTION

Write $e = (1, 1, \ldots, 1)^T \in \mathbb{R}^n$. We have the decomposition

$$A_n(a) = (1-a)I + a\, ee^T,$$

because every entry of ee^T equals 1.

1. Note that $ee^T e = (e^T e)\, e = n\, e$, so

$$A_n(a)e = (1-a)e + a\, ee^T e = (1-a)e + an\, e = \big(1 + (n-1)a\big)e,$$

Thus e is an eigenvector with eigenvalue $\lambda_1 = 1 + (n-1)a$.

Now let $x \in \mathbb{R}^n$ satisfy $e^T x = 0$. Then $ee^T x = e(e^T x) = 0$, so

$$A_n(a)x = (1-a)x + a \cdot 0 = (1-a)x.$$

Consequently, every $x \in e^\perp \setminus \{0\}$ is an eigenvector with eigenvalue $\lambda_2 = 1 - a$. Since $\mathbb{R}^n = \operatorname{span}\{e\} \oplus e^\perp$, these are all eigenvalues. If $a \neq 0$, then $\lambda_1 \neq \lambda_2$, so

$$\begin{aligned} &\lambda_1 = 1 + (n-1)a \quad \text{with algebraic and geometric multiplicity } 1, \\ &\lambda_2 = 1 - a \qquad\quad \text{with algebraic and geometric multiplicity } n-1. \end{aligned}$$

If $a = 0$, then $A_n(0) = I$, so its only eigenvalue is 1, with algebraic and geometric multiplicity n.

2. The matrix $A_n(a)$ is real symmetric and hence Hermitian. It is therefore positive definite exactly when all its eigenvalues are strictly positive, and positive semidefinite exactly when all its eigenvalues are nonnegative. Thus:

$$A_n(a) \succ 0 \iff \begin{cases} 1-a > 0, \\ 1 + (n-1)a > 0, \end{cases} \iff -\frac{1}{n-1} < a < 1,$$

and

$$A_n(a) \succeq 0 \iff \begin{cases} 1-a \geq 0, \\ 1 + (n-1)a \geq 0, \end{cases} \iff -\frac{1}{n-1} \leq a \leq 1.$$

3. If $a = 1$, then $\lambda_2 = 0$ has multiplicity $n-1$, and $\lambda_1 = n$. Hence

$$\operatorname{rank}(A_n(1)) = 1, \qquad \ker(A_n(1)) = e^\perp = \{x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 0\}.$$

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EXERCISE 13.4 CONTINUED

If $a = -\frac{1}{n-1}$, then $\lambda_1 = 0$ has multiplicity 1, and $\lambda_2 = 1 - a = \frac{n}{n-1} > 0$ has multiplicity $n-1$. Hence

$$\operatorname{rank}\Big(A_n\big(-\tfrac{1}{n-1}\big)\Big) = n-1, \qquad \ker\Big(A_n\big(-\tfrac{1}{n-1}\big)\Big) = \operatorname{span}\{e\}.$$

Now turn to $H = B^*B + I$.

(4) Let $x \in \mathbb{C}^n$. Then

$$x^*Hx = x^*(B^*B)x + x^*Ix = \|Bx\|^2 + \|x\|^2.$$

Hence $x^*Hx \geq 0$ for every x, and $x^*Hx > 0$ for every $x \neq 0$, since $\|x\|^2 > 0$. Thus $H \succ 0$, so H is invertible.

(5) From $H = I + B^*B$, we obtain $H - I = B^*B \succeq 0$; thus $x^*Hx \geq x^*Ix = \|x\|^2$ for every x. Let $\lambda_{\min}$ be the smallest eigenvalue of H. The preceding inequality gives $\lambda_{\min} \geq 1$. Every eigenvalue of H^{-1} is therefore at most 1, so for every x,

$$x^*H^{-1}x \leq \|x\|^2.$$

For the determinant inequality, write the eigenvalues of H as $\lambda_1, \ldots, \lambda_n$. Since $H \succ 0$, we have $\lambda_i > 0$; since $\lambda_{\min} \geq 1$, in fact $\lambda_i \geq 1$ for every i. Hence

$$\det(H) = \prod_{i=1}^{n} \lambda_i \geq 1.$$

Equality holds if and only if $\lambda_i = 1$ for every i: equivalently, $H = I$, or $B^*B = 0$. Finally, $B^*B = 0$ implies $\|Bx\|^2 = x^*B^*Bx = 0$ for every x. Thus $Bx = 0$ for all x, so $B = 0$.

V-I-C-T-O-R-Y!

EXERCISE 13.5 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ be symmetric matrices.

1. Please prove that there is an orthogonal matrix Q for which both Q^TAQ and Q^TBQ are diagonal if and only if $AB = BA$.
2. Please find such a matrix for

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 6 \end{pmatrix},$$

and also give the diagonal matrix representing AB in the same basis.

WORKED SOLUTION

1. If $Q^TAQ = D_A$ and $Q^TBQ = D_B$ are diagonal, then $A = QD_AQ^T$ and $B = QD_BQ^T$.

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EXERCISE 13.5 CHALLENGE CONTINUED

Since diagonal matrices commute,

$$AB = QD_AD_BQ^T = QD_BD_AQ^T = BA.$$

Conversely, suppose $AB = BA$. By the spectral theorem, $\mathbb{R}^n$ is the orthogonal direct sum of the eigenspaces of A. If $x \in E_\lambda(A)$, then

$$A(Bx) = B(Ax) = B(\lambda x) = \lambda Bx,$$

so $Bx \in E_\lambda(A)$. Thus B preserves every eigenspace of A. On each such space, the restriction of B is self-adjoint for the induced inner product: for all x, y in that space,

$$\langle Bx, y\rangle = \langle x, By\rangle,$$

because $B^T = B$. By the spectral theorem on that space, choose an orthonormal basis of eigenvectors of B. Each vector is also an eigenvector of A belonging to λ. Combining these bases over all eigenvalues of A gives an orthonormal basis of $\mathbb{R}^n$ common to both matrices. Placing it in the columns of Q gives the required diagonalizations, including the case of repeated eigenvalues.

2. Choose

$$q_1 = \frac{1}{\sqrt{2}}(1,1,0)^T, \qquad q_2 = \frac{1}{\sqrt{2}}(1,-1,0)^T, \qquad q_3 = (0,0,1)^T.$$

Their norms are 1 and their pairwise inner products are 0. Furthermore,

$$Aq_1 = -q_1, \quad Aq_2 = -q_2, \quad Aq_3 = 4q_3, \qquad Bq_1 = 7q_1, \quad Bq_2 = 3q_2, \quad Bq_3 = 6q_3.$$

Thus, for

$$Q = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

we obtain

$$Q^TAQ = \operatorname{diag}(-1,-1,4), \qquad Q^TBQ = \operatorname{diag}(7,3,6),$$
$$Q^TABQ = \operatorname{diag}(-7,-3,24).$$

Victory!

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EXERCISE 13.6 CHALLENGE

Let $A \in M_n(\mathbb{R})$ be symmetric and positive definite, meaning $x^T Ax > 0$ for every $x \neq 0$. Please prove that there is a unique symmetric positive definite matrix B satisfying $B^2 = A$.

WORKED SOLUTION

The spectral theorem gives an orthogonal matrix Q with

$$A = QDQ^T, \qquad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Every λ_j is positive, because a corresponding unit eigenvector q_j satisfies $\lambda_j = q_j^T Aq_j > 0$. Define

$$B = Q \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})Q^T.$$

It is symmetric and satisfies $B^2 = A$. For $x \neq 0$, put $y = Q^T x \neq 0$. Then $x^T Bx = \sum_j \sqrt{\lambda_j} y_j^2 > 0$. For uniqueness, let C be another symmetric positive definite square root. Since $CA = C^3 = AC$, it preserves every eigenspace $E_\lambda(A)$. Its restriction to that eigenspace is symmetric and positive definite, so it has an orthonormal eigenbasis. For a vector with $Cv = \mu v$, positivity gives $\mu > 0$, and

$$\mu^2 v = C^2 v = Av = \lambda v.$$

Thus $\mu = \sqrt{\lambda}$. On each eigenspace of A, both B and C act as multiplication by $\sqrt{\lambda}$. These eigenspaces span $\mathbb{R}^n$, proving $C = B$.

And we're done. Victory!

13.3 Normality and Unitary Diagonalization

NOTATION AND CONVENTIONS

The following exercises are over $\mathbb{C}$, and $A^* = \overline{A}^T$ is the conjugate transpose. A matrix U is unitary when $U^*U = UU^* = I$. A unitary diagonalization has the form $U^*AU = D$ with D diagonal. The inner product is linear in its first variable.

PROPOSITION 13.2

A matrix $A \in M_n(\mathbb{C})$ is unitarily diagonalizable if and only if it is normal, meaning $AA^* = A^*A$.

Proof.
If $A = UDU^*$, then $A^* = UD^*U^*$, so

$$AA^* = UDD^*U^*, \qquad A^*A = UD^*DU^*.$$

Diagonal matrices commute, giving $AA^* = A^*A$. Conversely, suppose A is normal, and set

$$H = \frac{A + A^*}{2}, \qquad K = \frac{A - A^*}{2i}.$$

We have $H^* = H$ and

$$K^* = \frac{-1}{2i}(A^* - A) = K,$$

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so both are Hermitian. Compute both products:

$$HK = \frac{1}{4i}\big(A^2 - AA^* + A^*A - (A^*)^2\big),$$
$$KH = \frac{1}{4i}\big(A^2 + AA^* - A^*A - (A^*)^2\big).$$

Normality gives $HK = KH$. By the Hermitian spectral theorem, the space is the orthogonal direct sum of the eigenspaces of H. If $Hx = \lambda x$, then

$$H(Kx) = K(Hx) = \lambda Kx,$$

so K preserves each eigenspace of H. Its restriction to each such space is Hermitian, since $\langle Kx, y\rangle = \langle x, Ky\rangle$ there. Choose an orthonormal eigenbasis of K in each space. Their union is a common orthonormal eigenbasis of H, K, and hence of $A = H + iK$. Its basis matrix U is unitary and satisfies $U^*AU = D$ diagonal. For $n = 0$, the empty basis proves the statement immediately.

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EXERCISE 13.7 CHALLENGE

Consider

$$A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}.$$

Please compute A^2. Please find a diagonalization $P^{-1}AP = D$, and please determine whether a unitary diagonalization also exists. Please determine whether $A^2 = I$ together with all eigenvalues on the unit circle forces A to be unitary.

WORKED SOLUTION

Direct multiplication gives

$$A^2 = \begin{pmatrix} 1 & 2-2 \\ 0 & 1 \end{pmatrix} = I_2.$$

Eigenvectors for $1, -1$ are $(1,0)^T$ and $(-1,1)^T$, respectively. Thus take

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Then

$$AP = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad P^{-1}AP = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

However,

$$A^*A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \qquad AA^* = \begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix}.$$

They differ, so A is not normal and has no unitary diagonalization. In particular, $A^*A \neq I_2$, so A is not unitary despite $A^2 = I_2$ and its eigenvalues $1, -1$ lying on the unit circle. Its eigendirections are not orthogonal: $\langle (1,0), (-1,1) \rangle = -1$.

You know what that means? V-I-C-T-O-R-Y!

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EXERCISE 13.8 CHALLENGE

Let $U \in M_n(\mathbb{C})$ be unitary, with $n \geq 1$.

1. Please prove that there is a unitary matrix S satisfying $S^2 = U$.
2. If U has n distinct eigenvalues, please find the number of unitary matrices S satisfying $S^2 = U$.
3. Please construct infinitely many distinct unitary matrices satisfying $S^2 = -I_2$.

WORKED SOLUTION

1. A unitary matrix is normal, so $U = V \operatorname{diag}(\lambda_1, \ldots, \lambda_n)V^*$ with V unitary. If $Uq_j = \lambda_j q_j$ and $q_j \neq 0$, then
$$\|q_j\| = \|Uq_j\| = |\lambda_j|\|q_j\|,$$
so $|\lambda_j| = 1$. Write $\lambda_j = e^{i\theta_j}$, and set
$$S = V \operatorname{diag}(e^{i\theta_1/2}, \ldots, e^{i\theta_n/2})V^*.$$
Every diagonal entry has modulus 1, giving $S^*S = SS^* = I$ and $S^2 = U$.

2. Every square root commutes with U, since $SU = S^3 = US$. Thus S preserves every eigenspace of U. Distinct eigenvalues make these spaces one-dimensional, so in the same eigenbasis
$$S = V \operatorname{diag}(\mu_1, \ldots, \mu_n)V^*, \qquad \mu_j^2 = \lambda_j.$$
Each nonzero λ_j has exactly two distinct complex square roots, both of modulus 1. The choices are independent, giving exactly 2^n such matrices.

3. For $0 \leq t < \pi$, let $v_t = (\cos t, \sin t)^T$ and let $P_t = v_t v_t^T$ be the orthogonal projection onto its span. We have $P_t^2 = P_t = P_t^*$. Set
$$S_t = i(2P_t - I_2) = i\begin{pmatrix} \cos(2t) & \sin(2t) \\ \sin(2t) & -\cos(2t) \end{pmatrix}.$$
Since $(2P_t - I_2)^2 = 4P_t^2 - 4P_t + I_2 = I_2$,
$$S_t^2 = -I_2, \qquad S_t^* S_t = (-i)(i)(2P_t - I_2)^2 = I_2.$$
The lines $\operatorname{span}\{v_t\}$ are distinct for $t \in [0, \pi)$, so their projections P_t and the matrices S_t are distinct. This gives an infinite family.

Another one down. Victory!

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EXERCISE 13.9 CHALLENGE

Let $A, B \in M_n(\mathbb{C})$ be normal.

1. Please prove that there is a unitary Q for which both Q^*AQ and Q^*BQ are diagonal if and only if $AB = BA$.
2. Please find such a simultaneous diagonalization for

$$A = \operatorname{diag}(1+i, 1+i, -2i), \qquad B = \begin{pmatrix} 2 & i & 0 \\ -i & 2 & 0 \\ 0 & 0 & 1+i \end{pmatrix}.$$

Please also compute Q^*ABQ.

WORKED SOLUTION

1. If $A = QD_AQ^*$ and $B = QD_BQ^*$, then

$$AB = QD_AD_BQ^* = QD_BD_AQ^* = BA.$$

Conversely, suppose $AB = BA$. Unitarily diagonalize $A = VDV^*$, and set $C = V^*BV$. Entrywise, $DC = CD$ gives

$$(\lambda_i - \lambda_j)c_{ij} = 0.$$

Thus C is block diagonal, with one block per distinct eigenvalue of A. Normality of B gives $CC^* = C^*C$, so every block is normal. Unitarily diagonalize each block, and combine the changes into a block-diagonal unitary matrix W. Then W^*CW is diagonal, while $W^*DW = D$, since each block of D is scalar. The matrix $Q = VW$ gives both diagonalizations.

2. The diagonal matrix A is normal. Both blocks of B are normal: the first is Hermitian and the second is scalar. The matrices commute because the first block of A is $(1+i)I_2$. Choose

$$q_1 = \frac{1}{\sqrt{2}}(i, 1, 0)^T, \quad q_2 = \frac{1}{\sqrt{2}}(-i, 1, 0)^T, \quad q_3 = (0, 0, 1)^T.$$

Their norms are 1, and $\langle q_1, q_2 \rangle = (i \cdot i + 1)/2 = 0$; also q_3 is orthogonal to both. Direct multiplication gives

$$Bq_1 = 3q_1, \qquad Bq_2 = q_2, \qquad Bq_3 = (1+i)q_3.$$

For $Q = (q_1\ q_2\ q_3)$, we obtain

$$\begin{aligned} Q^*AQ &= \operatorname{diag}(1+i, 1+i, -2i), \\ Q^*BQ &= \operatorname{diag}(3, 1, 1+i), \\ Q^*ABQ &= \operatorname{diag}(3+3i, 1+i, 2-2i). \end{aligned}$$

We've got it. Victory!

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EXERCISE 13.10 CHALLENGE

Let $A \in M_n(\mathbb{C})$ be normal, and suppose $\|A^2x\| = \|Ax\|$ for every $x \in \mathbb{C}^n$. Please prove that A^*A is an orthogonal projection matrix and that $\operatorname{Im} A = (\ker A)^\perp$. Please prove that the restriction of A to $\operatorname{Im} A$ is unitary. Please check whether normality can be dropped from the conclusion about A^*A.

WORKED SOLUTION

Write $A = UDU^*$ in an orthonormal eigenbasis. For a unit eigenvector q_j with eigenvalue λ_j, the assumption gives

$$|\lambda_j|^2 = \|A^2 q_j\| = \|Aq_j\| = |\lambda_j|.$$

Thus $|\lambda_j|$ is 0 or 1. With $\varepsilon_j = |\lambda_j|^2$,

$$A^*A = U \operatorname{diag}(\varepsilon_1, \ldots, \varepsilon_n) U^*, \qquad \varepsilon_j \in \{0, 1\}.$$

This matrix is Hermitian and satisfies $(A^*A)^2 = A^*A$, so it is an orthogonal projection. The kernel of A is spanned by the eigenvectors with $\lambda_j = 0$, and its image by the remaining eigenvectors: for $\lambda_j \neq 0$,

$$q_j = A(\lambda_j^{-1} q_j).$$

Orthonormality gives $\operatorname{Im} A = (\ker A)^\perp$. On the image, the restriction is diagonal with all entries of modulus 1, hence unitary. If the image is the zero space, the empty basis gives the assertion. Without normality, take

$$C = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}.$$

Then $C^2 = C$, so $\|C^2x\| = \|Cx\|$ for every x. However,

$$C^*C = \begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix}, \qquad (C^*C)^2 = 10C^*C \neq C^*C.$$

Thus C^*C is not a projection.

That does it. Victory!

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EXERCISE 13.11 CHALLENGE

For a matrix $M = (m_{jk}) \in M_n(\mathbb{C})$, define its Frobenius norm by

$$\|M\|_F^2 = \sum_{j,k=1}^{n} |m_{jk}|^2 = \mathrm{tr}(M^* M).$$

Let $A \in M_n(\mathbb{C})$ be normal and invertible.

1. Please find the unitary matrix U minimizing $\|A - U\|_F$, and please prove that it is unique.
2. Please compute it and the minimum distance for

$$A = \frac{1}{2}\begin{pmatrix} -3+2i & 3+2i \\ 3+2i & -3+2i \end{pmatrix}.$$

Hint: please use an orthonormal eigenbasis of A, and please use $|w_{jj}| \le 1$ for every unitary matrix W.

WORKED SOLUTION

The Frobenius norm is unchanged by multiplication by unitary matrices. For example, if V is unitary,

$$\|V^* M V\|_F^2 = \mathrm{tr}(V^* M^* V V^* M V) = \mathrm{tr}(V^* M^* M V) = \mathrm{tr}(M^* M).$$

The last step uses $\mathrm{tr}(XY) = \mathrm{tr}(YX)$, obtained by summing the entries $\sum_{j,k} x_{jk} y_{kj}$ in either order. Write $A = VDV^*$ with $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ and every $\lambda_j \neq 0$. For each unitary U, set $W = V^* U V$, which is also unitary. Then

$$\begin{aligned} \|A - U\|_F^2 &= \|D - W\|_F^2 \\ &= \sum_j |\lambda_j|^2 + n - 2\,\mathrm{Re} \sum_j \lambda_j \overline{w_{jj}} \\ &\ge \sum_j |\lambda_j|^2 + n - 2\sum_j |\lambda_j| \\ &= \sum_j (|\lambda_j| - 1)^2. \end{aligned}$$

Here $\sum_{j,k} |w_{jk}|^2 = n$ because each column of W is a unit vector; in particular $|w_{jj}| \le 1$. Equality is attained by

$$W_0 = \mathrm{diag}\left(\frac{\lambda_1}{|\lambda_1|}, \ldots, \frac{\lambda_n}{|\lambda_n|}\right), \qquad U_0 = V W_0 V^*.$$

For uniqueness, every bound $\mathrm{Re}(\lambda_j \overline{w_{jj}}) \le |\lambda_j|$ must be an equality. As $\lambda_j \neq 0$, this forces $w_{jj} = \lambda_j / |\lambda_j|$, hence $|w_{jj}| = 1$. The squared moduli in that row sum to 1, so every other entry in the row is zero. Thus $W = W_0$ and $U = U_0$. In the numerical case, take

$$V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad V^* A V = \mathrm{diag}(2i, -3).$$

Therefore

$$U_0 = V\,\mathrm{diag}(i, -1) V^* = \frac{1}{2}\begin{pmatrix} -1+i & 1+i \\ 1+i & -1+i \end{pmatrix},$$

and the distance is $\sqrt{(2-1)^2 + (3-1)^2} = \sqrt{5}$.

All wrapped up. Victory!

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13.4 Reference Table: Matrices and Linear Maps

NOTATION AND CONVENTIONS

In this table, V, W are finite-dimensional spaces over $\mathbb{F}$, with dimensions n, m and ordered bases B, C. For $T : V \to W$, write $A = [T]_C^B \in M_{m\times n}(\mathbb{F})$. Rows involving a subspace U assume $U \leq V$, and $[U]_B = \{[u]_B : u \in U\}$ denotes its coordinate set. Square-matrix properties refer to $T : V \to V$, using the same basis in domain and codomain. Inner-product rows are over $\mathbb{R}$ or $\mathbb{C}$; identities involving A^* use orthonormal bases, except for the row explicitly introducing the matrix G. The symbol I denotes an identity, and 0 denotes a zero vector, map or matrix as appropriate. The symbols ker, Im, Row and Col denote kernel, image, row space and column space. For the zero space the basis is empty, rank and nullity are 0, and the empty matrix has determinant 1 and trace 0.

Concept	Maps and spaces	Matrix form
Vector and coordinates	$v = \sum_j x_j b_j$; the coordinates are its coefficients in the basis B.	$[v]_B = (x_1, \ldots, x_n)^T$.
Linear combination	A vector $\sum_j \alpha_j v_j$ with scalars $\alpha_j \in \mathbb{F}$.	If the columns of M are $[v_j]_B$, then $[\sum_j \alpha_j v_j]_B = M\alpha$.
Span	$\operatorname{span}\{v_j\}$ is the set of all linear combinations of the list.	The column space of its coordinate matrix.
Linear independence	$\sum_j \alpha_j v_j = 0$ forces every $\alpha_j = 0$.	The columns of M are independent exactly when $\ker M = \{0\}$.
Basis	An independent list spanning the space.	The coordinate matrix of a basis of $\mathbb{F}^n$ is square and invertible.
Dimension	$\dim V$ is the number of vectors in any basis of V.	$\dim \mathbb{F}^n = n$; $\dim M_{m\times n}(\mathbb{F}) = mn$.
Linear map	$T(\alpha v + \beta w) = \alpha T(v) + \beta T(w)$.	$[T(v)]_C = A[v]_B$.
Representing matrix	Column j gives the coordinates of $T(b_j)$ in C.	$A = [T]_C^B$. $A_{\cdot j} = [T(b_j)]_C$.
Composition	$S \circ T$ applies T first and then S.	$[S \circ T]_D^B = [S]_D^C [T]_C^B$; the intermediate bases must agree.
Kernel	$\ker T = \{v \in V : T(v) = 0\}$.	$[\ker T]_B = \ker A$; the solutions of $Ax = 0$.
Image	$\operatorname{Im} T = \{T(v) : v \in V\}$.	$[\operatorname{Im} T]_C = \operatorname{Col} A$.
Rank	$\operatorname{rank} T = \dim \operatorname{Im} T$.	$\operatorname{rank} A = \dim \operatorname{Col} A = \dim \operatorname{Row} A$.
Nullity	$\operatorname{nullity} T = \dim \ker T$.	$\operatorname{nullity} A = n - \operatorname{rank} A$.
Rank-nullity	$\dim \ker T + \dim \operatorname{Im} T = n$.	The number of free variables plus the rank equals the number of columns.
Injectivity	$T(v) = T(w)$ implies $v = w$; equivalently $\ker T = \{0\}$.	The columns of A are independent; $\operatorname{rank} A = n$.
Surjectivity	$\operatorname{Im} T = W$.	The columns of A span $\mathbb{F}^m$; $\operatorname{rank} A = m$.
Isomorphism	A linear map that is both injective and surjective.	$m = n$ and A is invertible.
Inverse	$T^{-1}T = I_V$ and $TT^{-1} = I_W$.	$[T^{-1}]_B^C = A^{-1}$; $AA^{-1} = A^{-1}A = I$. A square matrix is singular when it is not invertible, equivalently $\det A = 0$.

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Concept	Maps and spaces	Matrix form
Row space	The subspace spanned by the rows, viewed as vectors in $\mathbb{F}^n$.	Row $A \leq \mathbb{F}^n$; row operations preserve it.
Column space	The subspace spanned by the columns, in $\mathbb{F}^m$.	Col $A = \mathrm{Im}(x \mapsto Ax)$.
Row equivalence	A sequence of invertible row operations; applied to an augmented system, they preserve its solutions.	$B = EA$ with E invertible, a product of elementary matrices.
Change of basis	P converts B' coordinates into B coordinates.	$[v]_B = P[v]_{B'}$; column j is $[b'_j]_B$.
Similarity	Two matrices of the same endomorphism $T : V \to V$ in different bases.	$D = P^{-1}AP$ with P invertible.
Trace	A change of basis preserves the sum of diagonal entries.	$\mathrm{tr}\, A = \sum_j a_{jj}$; $\mathrm{tr}(XY) = \mathrm{tr}(YX)$ when both products are defined and square.
Determinant	A scalar attached to an endomorphism, independent of the chosen basis.	$\det A$; $\det(AB) = \det A \det B$. Over $\mathbb{R}$, its absolute value is the volume scale factor.
Minor and cofactor	For $n \geq 1$, $A_{\widehat{i},\widehat{j}}$ is obtained by deleting row i and column j.	$C_{ij} = (-1)^{i+j} \det A_{\widehat{i},\widehat{j}}$.
Classical adjugate	The transpose of the cofactor matrix; here $n \geq 1$.	$\mathrm{adj}(A) = (C_{ij})^T$; $A\,\mathrm{adj}(A) = \det(A)I$.
Eigenvalue and eigenvector	$T(v) = \lambda v$ with $v \neq 0$.	$Ax = \lambda x$ with $x \neq 0$; $\det(A - \lambda I) = 0$.
Eigenspace	$E_\lambda(T) = \ker(T - \lambda I_V)$.	$E_\lambda(A) = \ker(A - \lambda I_n)$.
Characteristic polynomial	Independent of basis; its roots in the field are the eigenvalues.	$p_A(t) = \det(tI_n - A)$; a monic polynomial of degree n.
Algebraic multiplicity	The multiplicity of λ as a root of the characteristic polynomial.	If $p_A(t) = (t - \lambda)^r q(t)$ and $q(\lambda) \neq 0$, the multiplicity is r.
Geometric multiplicity	The dimension of $E_\lambda(T)$.	$\dim \ker(A - \lambda I)$; between 1 and the algebraic multiplicity for an eigenvalue λ.
Diagonalization	Existence of a basis of eigenvectors of T.	$P^{-1}AP = D$ diagonal. Equivalent to splitting of the characteristic polynomial over the field and equality of both multiplicities at every eigenvalue.
Cayley-Hamilton	Substituting the endomorphism into its characteristic polynomial gives the zero map.	$p_A(A) = 0$; a constant term c becomes cI_n.
Minimal polynomial	The monic polynomial of least degree annihilating the endomorphism.	$m_A(A) = 0$ and $m_A \mid p_A$; for $n \geq 1$.
Nilpotence	$T^k = 0$ for some $k \geq 1$.	$A^k = 0$ for some $k \geq 1$; every eigenvalue is zero.

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Concept	Maps and spaces	Matrix form
Idempotence	$T^2 = T$; the map acts as the identity on its image.	$A^2 = A$; projection onto the image along the kernel.
Involution	$T^2 = I_V$; applying the map twice recovers the input.	$A^2 = I_n$; $A^{-1} = A$.
Zero and identity	The zero map sends every vector to zero; I_V fixes every vector.	$0_{m\times n}$ has only zero entries. The diagonal of I_n consists of ones, and all other entries are zero.
Diagonal matrix	In the basis B, $T(b_j) = d_j b_j$ for every j.	$D = \operatorname{diag}(d_1, \ldots, d_n)$; every off-diagonal entry is zero.
Triangular matrix	For $1 \le k \le n$, upper triangular preserves every $\operatorname{span}\{b_1, \ldots, b_k\}$; lower triangular preserves every $\operatorname{span}\{b_k, \ldots, b_n\}$.	Upper: $a_{ij} = 0$ for $i > j$. Lower: $a_{ij} = 0$ for $i < j$.
Scalar matrix	$T(v) = \lambda v$ for every $v \in V$, for a fixed $\lambda \in \mathbb{F}$.	$A = \lambda I_n$. Its matrix is the same in every basis of V.
Invariant subspace	$U \le V$ satisfies $T(U) \subseteq U$. A basis of U can be extended to a basis of V.	In such a basis, partitioned as $k + (n-k)$ with $k = \dim U$: $\begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix}$.
Elementary matrix	Represents one elementary row operation. A row-scaling factor must be nonzero; every such operation is invertible.	E is obtained from I_m by one operation. The product EA applies that operation to the rows of A.
Permutation matrix	Permutes the basis vectors: $T(e_j) = e_{\sigma(j)}$, where σ permutes $\{1, \ldots, n\}$.	Every row and column has one entry equal to 1 and all others zero. $P^{-1} = P^T$.
Transpose	Interchange the row and column indices.	$(A^T)_{ij} = a_{ji}$; $(AB)^T = B^T A^T$.
Conjugate transpose	Over $\mathbb{R}$ or $\mathbb{C}$, transpose and conjugate every entry.	$(A^*)_{ij} = \overline{a_{ji}}$; $(AB)^* = B^* A^*$.
Adjoint map	$\langle Tv, w\rangle_W = \langle v, T^* w\rangle_V$ for all v, w.	In orthonormal bases: $[T^*]_B^C = A^*$.
Inner product	Linear in its first variable, conjugate symmetric and positive definite.	In an orthonormal basis: $\langle v, w\rangle = [w]^*[v]$.
Inner-product matrix	In a basis $B = (b_j)$, set $G_{ij} = \langle b_j, b_i\rangle$. Then $\langle v, w\rangle = [w]_B^* G [v]_B$.	In the same basis, for $T : V \to V$: $[T^*]_B = G^{-1} A^* G$.
Induced norm	$\|v\| = \sqrt{\langle v, v\rangle}$.	In an orthonormal basis: $\|v\|^2 = \sum_j |x_j|^2$.
Orthogonality	$v \perp w$ means $\langle v, w\rangle = 0$.	In an orthonormal basis: $[w]^*[v] = 0$.
Orthonormal basis	$\langle e_i, e_j\rangle = \delta_{ij}$; $\delta_{ij} = 1$ if $i = j$, and zero otherwise.	The basis matrix Q satisfies $Q^* Q = I$.
Orthogonal complement	$U^\perp = \{v : \langle v, u\rangle = 0 \text{ for all } u \in U\}$.	If the columns of M span U in orthonormal coordinates, then $[U^\perp]_B = \ker M^*$.

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Concept	Maps and spaces	Matrix form
Orthogonal projection	$P_U v \in U$ and $v - P_U v \in U^\perp$.	$P^2 = P = P^*$; if the columns of Q are an orthonormal basis of U, then $P = QQ^*$.
Gram-Schmidt	Subtract projections onto preceding directions and normalize each nonzero residual.	For independent input: $w_k = v_k - \sum_{j<k} \langle v_k, e_j \rangle e_j$; $e_k = w_k / \|w_k\|$.
Linear isometry	$\langle Tv, Tw \rangle = \langle v, w \rangle$ for all v, w.	In orthonormal bases: $A^*A = I_n$, even when A is rectangular.
Unitarity	An invertible inner-product-preserving endomorphism over $\mathbb{C}$.	$A^*A = AA^* = I_n$; $A^{-1} = A^*$.
Orthogonal matrix	An invertible isometry of a real inner product space.	$A^TA = AA^T = I_n$; $A^{-1} = A^T$.
Self-adjoint and Hermitian	$T = T^*$; every eigenvalue is real.	$A = A^*$ in an orthonormal basis. Over $\mathbb{R}$: $A = A^T$.
Symmetry	The algebraic matrix condition $A^T = A$. Over $\mathbb{R}$, it represents self-adjointness in an orthonormal basis.	Over $\mathbb{C}$, $A^T = A$ differs from $A^* = A$.
Skew-adjointness	$T^* = -T$; over $\mathbb{C}$ every eigenvalue is purely imaginary.	$A^* = -A$ in an orthonormal basis.
Skew-symmetry	The matrix condition $A^T = -A$; in characteristic different from 2, the diagonal is zero.	Over $\mathbb{R}$, in an orthonormal basis, it represents skew-adjointness.
Normality	$TT^* = T^*T$.	$AA^* = A^*A$ in an orthonormal basis; over $\mathbb{C}$, equivalent to unitary diagonalizability.
Positive definiteness	$T = T^*$ and $\langle Tv, v \rangle > 0$ for all $v \neq 0$.	$A = A^*$ and $x^*Ax > 0$ for all $x \neq 0$; every eigenvalue is positive.
Positive semidefiniteness	$T = T^*$ and $\langle Tv, v \rangle \geq 0$ for all v.	$A = A^*$ and $x^*Ax \geq 0$ for all x; every eigenvalue is nonnegative.
Unitary diagonalization	Existence of an orthonormal eigenbasis in a complex inner product space.	$U^*AU = D$ diagonal with U unitary; equivalent to normality.
Orthogonal diagonalization	Existence of a real orthonormal eigenbasis.	$Q^TAQ = D$ real diagonal with Q orthogonal; equivalent to symmetry of $A \in M_n(\mathbb{R})$.
Frobenius norm	A norm on matrix space: the square root of the sum of squared moduli of all entries.	$\|A\|_F^2 = \sum_{i,j} \|a_{ij}\|^2 = \operatorname{tr}(A^*A)$.

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